

Package ‘MVCens’

March 3, 2026

Title Matrix-Variate Models with Censoring and Missing Data

Version 0.0.0.9000

Author Átila Correia [aut, cre], Victor Lachos [aut], Carlos Diniz [ctb]

Maintainer Átila Correia <atilapc1989@gmail.com>

Description Tools for generating and estimating matrix-variate Normal (MVN) and matrix-variate Skew-Normal (MVSN) models with or without interval censoring and missing data. The package provides functions to simulate uncensored and censored matrix-valued observations from MVN and MVSN distributions, as well as routines for likelihood evaluation. Estimation procedures include maximum likelihood for complete MVN data and EM-type algorithms for MVN and MVSN models under interval censoring or missingness. Core functionality covers the simulation of matrix-variate samples via the stochastic representation $X = M + A|Z| + V$, and the generation of censored and uncensored data for estimation.

License MIT + file LICENSE

Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.3

Imports LaplacesDemon,

Matrix,
matrixNormal,
mnormt,
MomTrunc,
mvtnorm,
stats,
tlrmvnmvt

Depends R (>= 3.5)

LazyData true

Contents

mv_ecm 2

rmatrix_censored	5
rmvsn	8
valle_dj_1920_1934	9

Index	11
--------------	-----------

mv_ecm	<i>ECM Algorithm for the MVN, MVSN, MVNC and MVSNC models</i>
--------	---

Description

This function provides a unified ECM-type estimation method for several matrix-variate distributions — MVN, MVSN, MVNC, and MVSNC — allowing for data sets that include interval-censored or missing values.

Usage

```
mv_ecm(  
  dist,  
  samples = NULL,  
  cc = NULL,  
  LS = NULL,  
  precision = 1e-07,  
  MaxIter = 50  
)
```

Arguments

dist	A character string specifying the distribution/model to be fitted. Options are: "MVN" – Matrix Variate Normal for Complete Data, "MVNC" – Censored Matrix Variate Normal, "MVSN" – Matrix Variate Skew-Normal for Complete Data, "MVSNC" – Censored Matrix Variate Skew-Normal.
samples	A 3D array of dimension $p \times q \times n$ containing the observed matrix-valued responses. For uncensored entries, the array stores the observed values; for censored entries, it stores the lower limits. This means that it stores the value -Inf when we are dealing with missing data or left censored data with no lower bound.
cc	A 3D indicator array with 1 for interval-censored or missing entries and 0 for observed values.
LS	A 3D array storing upper limits for interval-censored entries and 0 for non-censored entries. If we are dealing with missing data or right censored data with no upper bound, then the LS assumes the value +Inf.
precision	Convergence tolerance.
MaxIter	Maximum number of iterations.

Value

A list with components:

mu	Estimated location matrix ($p \times q$).
A	Estimated skewness matrix ($p \times q$). Returned only for MVSN and MVSNC models.
Sigma	Estimated row covariance matrix ($p \times p$).
Psi	Estimated column covariance matrix ($q \times q$).
loglik	Final log-likelihood value achieved by the ECM algorithm.
BIC	Bayesian Information Criterion.
iter	Number of ECM iterations performed.

Examples

```
#####
# Example 1: Fit an MVNC model to simulated 2x2 matrices
#####

set.seed(123)

# Dimensions: p = 2, q = 2, n = 20
p <- 2
q <- 2
n <- 20

# True parameters
M <- matrix(0, p, q)
A <- matrix(0, p, q)
U <- diag(p)
V <- diag(q)

# Simulated incomplete data
samples <- rmatrix_censored(n = n,
                           cens = 0.15,
                           Ind = 1,
                           M = M,
                           U = U,
                           V = V,
                           A = A,
                           dist = "Normal")

dados <- samples$X.cens
cc <- samples$cc
LS <- samples$LS

# Run ECM algorithm for censored MVN model
out <- mv_ecm("MVNC", samples = dados, cc = cc, LS = LS, MaxIter = 20)

# Inspect estimated parameters
out$mu
out$Sigma
```

```

out$Psi
out$loglik
out$BIC
out$iter

#####
# Example 2: Fit an MVN model to simulated 2x2 matrix-variate data
#####
set.seed(123)

# Dimensions: p = 2, q = 2, n = 20
p <- 2
q <- 2
n <- 20

# True parameters
M_true <- matrix(0, p, q)
Sigma_true <- diag(p)
Psi_true <- diag(q)

# Generate sample matrices from MVN(M_true, Sigma_true, Psi_true)
samples <- array(0, dim = c(p, q, n))
for (i in 1:n) {
  samples[, , i] <- LaplacesDemon::rmatrixnorm(
    M = M_true,
    U = Sigma_true,
    V = Psi_true
  )
}

# Run ECM algorithm for the Matrix-Variate Normal model
out <- mv_ecm("MVN", samples = samples, cc = NULL, LS = NULL, MaxIter = 20)

# Inspect fitted parameters
out$mu
out$Sigma
out$Psi
out$loglik
out$BIC
out$iter

#####
# Example 3: Fit an MVSN model to simulated 2x2 matrix-variate data
#####
p <- 3
q <- 2
n <- 20
M <- matrix(0, p, q)
A <- matrix(1, p, q)
U <- diag(p)
V <- diag(q)

# Generate simple test data

```

```

set.seed(123)
dados <- rmvsn(n = n, M = M, A = A, U = U, V = V)

out <- mv_ecm("MVSNC", samples = dados, cc = NULL, LS = NULL, MaxIter = 20)
out$mu
out$A
out$Sigma
out$Psi
out$loglik
out$BIC
out$iter

str(out)

#####
# Example 4: Fit an MVSNC model to simulated 2x2 matrix-variate data
#####

p <- 2
q <- 2
n <- 20
M <- matrix(0, p, q)
A <- matrix(1, p, q)
U <- diag(p)
V <- diag(q)

# Simulated incomplete data
samples <- rmatrix_censored(n = n, cens = 0.15, Ind = 1, M = M, U = U, V = V, A = A, dist = "SN")
dados <- samples$X.cens
cc <- samples$cc
LS <- samples$LS

out <- mv_ecm("MVSNC", samples = dados, cc = cc, LS = LS, MaxIter = 20)

out$mu
out$A
out$Sigma
out$Psi
out$loglik
out$BIC
out$iter

str(out)

```

Description

This function generates matrix-valued observations from either a Matrix Variate Normal (MVN) or Matrix Variate Skew Normal (MVSN) distribution, optionally applying missing or interval-censoring based on a quantile cutoff.

Usage

```
rmatrix_censored(
  n = n,
  cens = cens,
  Ind = 1,
  M = M,
  U = U,
  V = V,
  A = A,
  dist = "SN"
)
```

Arguments

<code>n</code>	Integer. Number of generated matrices.
<code>cens</code>	Numeric between 0 and 1. Censoring proportion used as the quantile cutoff. For example, <code>cens = 0.15</code> means censoring values below the 15 percent quantile.
<code>Ind</code>	Integer indicating the type of censoring: 1 = symmetric interval censoring (left and right), 2 = missing, 3 = mixed censoring (half missing values and half censored values), 4 = left censoring, 5 = right censoring.
<code>M</code>	Location matrix with dimensions p by q .
<code>U</code>	Row scale matrix with dimensions p by p .
<code>V</code>	Column scale matrix with dimensions q by q .
<code>A</code>	Skewness matrix with dimensions p by q . Used only when <code>dist = "SN"</code> .
<code>dist</code>	Character string indicating the distribution. Use "Normal" for MVN or "SN" for MVSN.

Details

Matrix-variate data are generated using the function `LaplaceDemon::rmatrixnorm`. When `dist = "SN"`, skewness is introduced by adding `A` multiplied by the absolute value of a standard normal draw.

Censoring is implemented using a quantile-based threshold controlled by the argument `cens`. The mechanism is specified by `Ind`.

Value

A list with components:

- `X.cens`: Censored data as an array of size p by q by n .
- `cc`: Censoring indicator array (1 means censored, 0 means observed).
- `LS`: Upper limits associated with censoring.

Examples

```

# Example: Generate censored matrix-variate data from
#           Normal and Skew-Normal models

set.seed(123)

# Dimensions: matrices of size p x q, sample size n
p <- 2
q <- 2
n <- 20

# Model parameters
M <- matrix(0, p, q)      # Location matrix
U <- diag(p)              # Row covariance
V <- diag(q)              # Column covariance
A <- matrix(1, p, q)      # Skewness matrix

# censoring proportion
cens <- 0.15

# -----
# 1) Generate censored MVN (Normal) matrix-variate data
# -----
out_normal <- rmatrix_censored(
  n = n,
  cens = cens,
  Ind = 1,      # interval-censored
  M = M,
  U = U,
  V = V,
  A = A,
  dist = "Normal"
)

# Extract components
X_normal <- out_normal$X.cens # true observations or lower censoring limits
cc_normal <- out_normal$cc    # censoring indicators
LS_normal <- out_normal$LS    # upper censoring limits or zero otherwise

# -----
# 2) Generate censored MVSN (Skew-Normal) matrix data
# -----
out_skew <- rmatrix_censored(
  n = n,
  cens = cens,
  Ind = 1,      # interval-censored
  M = M,
  U = U,
  V = V,
  A = A,
  dist = "SN"
)

```

```

)

# Extract components
X_skew <- out_skew$X.cens
cc_skew <- out_skew$cc
LS_skew <- out_skew$LS

# Display the first generated matrix
X_skew[, , 1]

```

rmvsn

*Generate Matrix Variate Skew Normal Samples (Complete Data)***Description**

Generates a sample of matrices following the matrix-variate skew-normal model $X = M + A|W| + \mathcal{V}$, where $W \sim N(0, 1)$ and $\mathcal{V} \sim \mathcal{MVN}_{p \times q}(0, U, V)$.

Usage

```
rmvsn(n = n, M = M, A = A, U = U, V = V)
```

Arguments

n	Integer. Number of matrices to generate.
M	Mean matrix of dimension $p \times q$.
A	Skewness matrix of dimension $p \times q$.
U	Row covariance matrix $p \times p$.
V	Column covariance matrix $q \times q$.

Details

To be more precise, this function simulates a sample of size n using the stochastic representation:

$$X_i = M + A|W_i| + \mathcal{V}_i$$

where

$$\mathcal{V}_i$$

is generated using `rmatrixnorm()`, which is provided by the **LaplacesDemon** package.

Value

A 3D array of generated samples with dimensions $p \times q \times n$.

Examples

```

p <- 3
q <- 2
n <- 20
M <- matrix(0, p, q)
A <- matrix(1, p, q)
U <- diag(p)
V <- diag(q)

samples <- rmvs(n = n, M = M, A = A, U = U, V = V)
dim(samples) # Should be 3 x 2 x 20

```

valle_dj_1920_1934	<i>Dow Jones dividends and divisor (quarterly), 1920–1934</i>
--------------------	---

Description

Quarterly Dow Jones (DJ) series used in the Valle application example. The dataset contains two variables observed for each quarter from 1920 to 1934: DJ dividends and the DJ divisor.

Usage

```
data(valle_dj_1920_1934)
```

Format

A numeric array with dimensions 4 x 2 x 15 and dimnames:

Quarter "Q1", "Q2", "Q3", "Q4".

Variable "DJ dividends" and "DJ divisor".

Year Character years from "1920" to "1934".

Details

The object is a 3D array indexed as [Quarter, Variable, Year]. Use `dimnames(valle_dj_1920_1934)` to see the labels.

Examples

```

data(valle_dj_1920_1934)

## Basic checks
dim(valle_dj_1920_1934)
dimnames(valle_dj_1920_1934)

## Extract the 1929 quarterly values (4 x 2 matrix)
valle_dj_1920_1934[, , "1929"]

```

```
## Extract DJ dividends for all quarters/years (4 x 15 matrix)
dividends <- valle_dj_1920_1934[, "DJ dividends", ]
head(dividends)

## Application example
# res <- mv_ecm("MVSN",
#               samples = valle_dj_1920_1934,
#               cc = NULL, LS = NULL,
#               precision = 1e-7, MaxIter = 50
# )
```

Index

* **datasets**

valle_dj_1920_1934, [9](#)

mv_ecm, [2](#)

rmatrix_censored, [5](#)

rmvsn, [8](#)

valle_dj_1920_1934, [9](#)